

EMPLOYEE ATTRITION ANALYSIS & PREDICTION

Data Analysis Project Report

IBM HR Analytics Dataset | Python | Machine Learning | Power BI

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1. Executive Summary

Employee attrition is one of the most costly challenges organisations face today. Replacing a single employee can cost between 50% to 200% of their annual salary. This project analyses the IBM HR Analytics Employee Attrition dataset to identify the key factors driving employee attrition and builds a predictive machine learning model to flag at-risk employees.

Metric	Value
Dataset	IBM HR Analytics Employee Attrition
Total Records Analysed	1,470 employees
Overall Attrition Rate	16.1%
Employees Who Left	237 (16.1%)
Employees Who Stayed	1,233 (83.9%)
Best ROC-AUC	0.807 (Logistic Regression)
Best Accuracy	85% (XGBoost)
Top Attrition Driver	Monthly Income
Tools Used	Python, Pandas, Seaborn, Scikit-learn, XGBoost, Power BI

2. Problem Statement

The HR department wants to understand why employees are leaving and identify which employees are most at risk. Key questions:

- What is the overall attrition rate?
- Which departments and job roles have the highest attrition?
- What are the key factors that drive an employee to leave?
- Can we predict which employees are likely to leave?
- What actions can HR take to reduce attrition?

3. Exploratory Data Analysis — Outputs

3.1 Overall Attrition Distribution

Out of 1,470 employees, 237 (16.1%) left the organisation and 1,233 (83.9%) stayed.

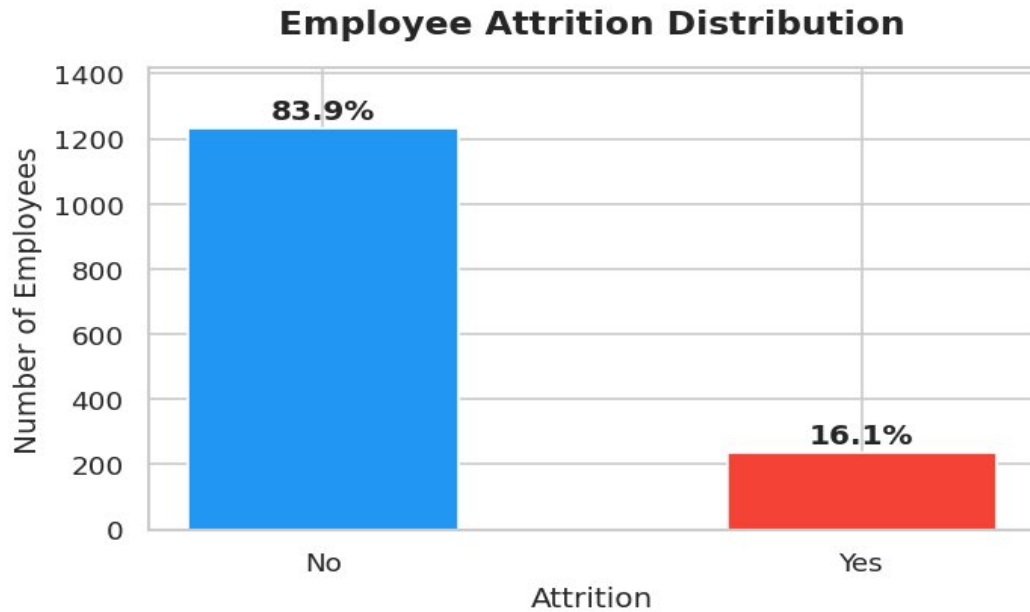


Figure 1: Employee Attrition Distribution — 16.1% attrition rate

3.2 Attrition by Department

Sales has the highest attrition rate at 20.6%, followed by Human Resources at 19.0%.

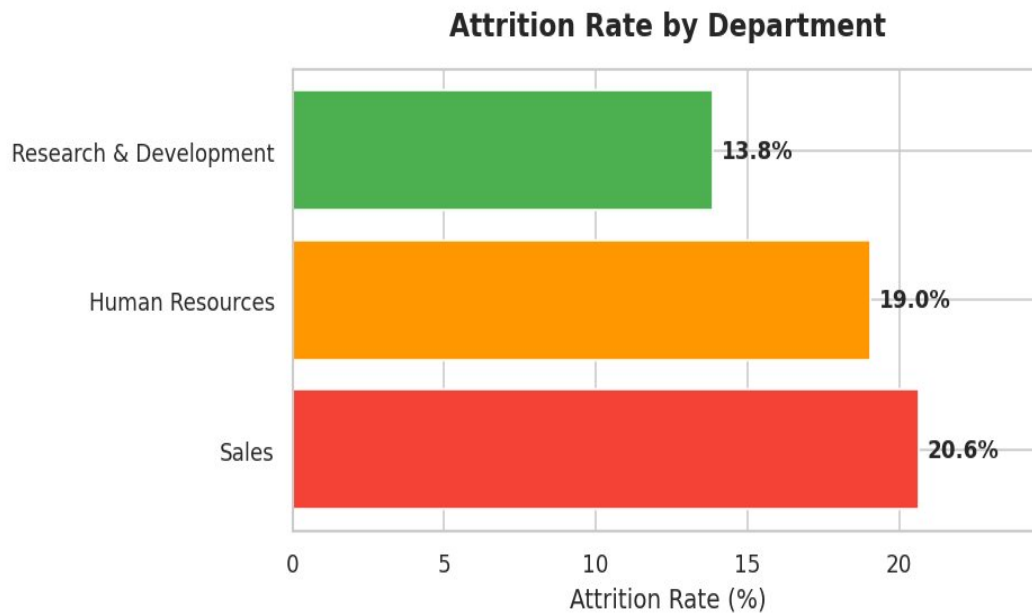


Figure 2: Attrition Rate by Department

3.3 Attrition by Age Group

Young employees aged 18-25 have the highest attrition rate at 34.8% — more than double any other age group.

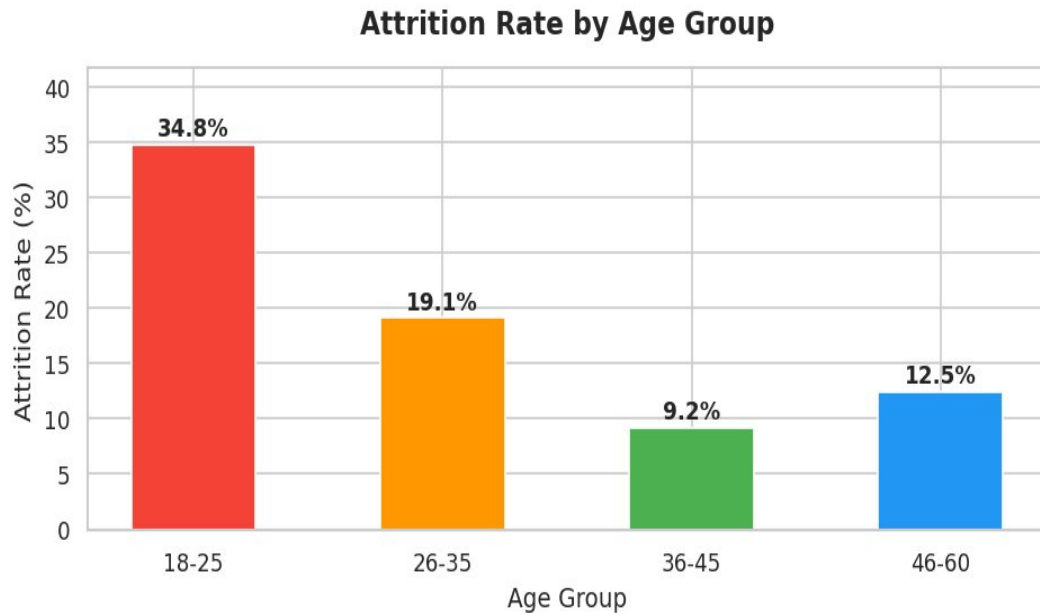


Figure 3: Attrition Rate by Age Group — 18-25 most at risk at 34.8%

3.4 Monthly Income Distribution

Employees who left were concentrated in lower income brackets, confirming salary as a key attrition driver.

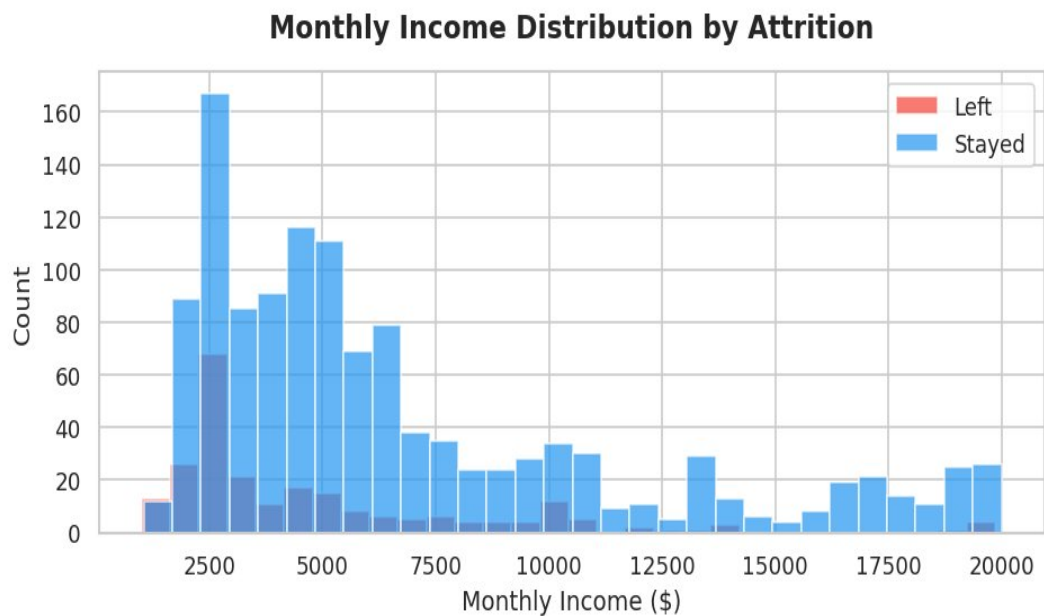


Figure 4: Monthly Income Distribution by Attrition Status

3.5 Attrition by Job Role

Sales Representatives have by far the highest attrition at 39.8%, nearly double the next highest role.

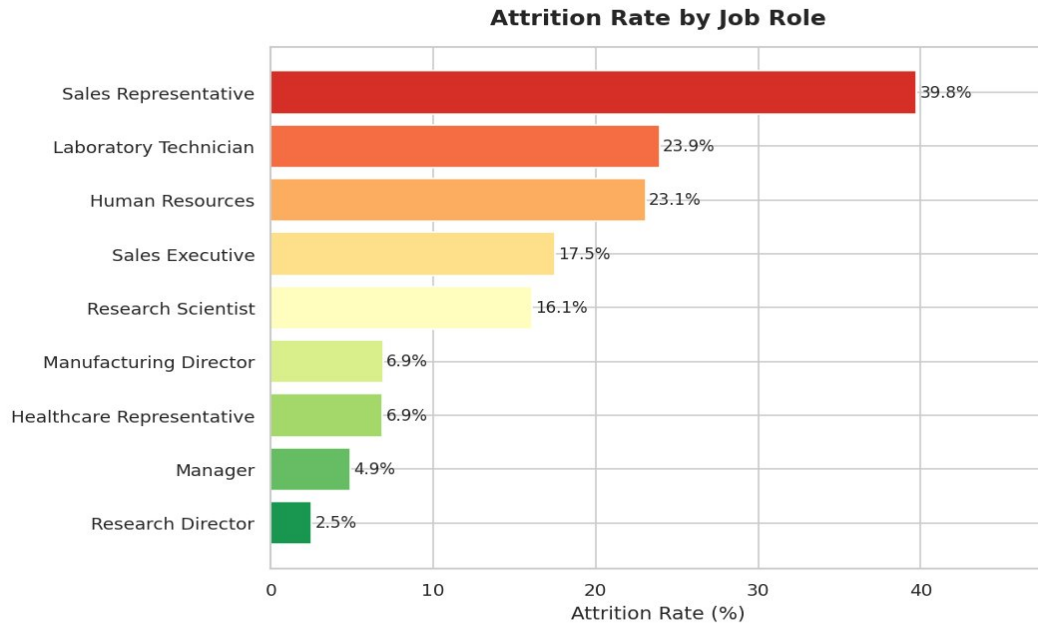


Figure 5: Attrition Rate by Job Role — Sales Representatives at 39.8%

3.6 Overtime vs Attrition

Employees doing overtime are 3x more likely to leave — 30.5% attrition vs 10.4% for those without overtime.

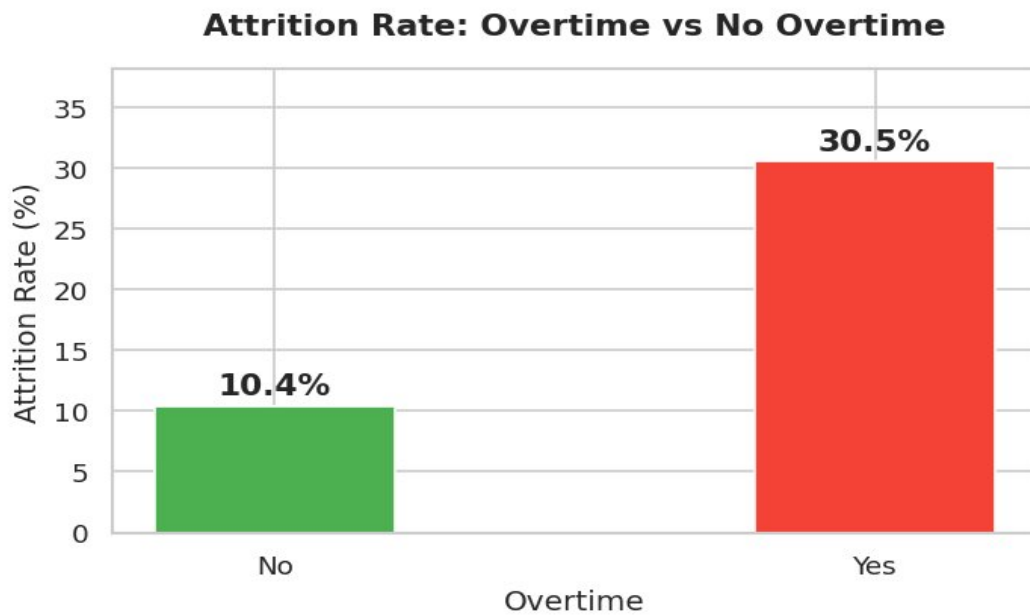


Figure 6: Overtime vs No Overtime Attrition Rate

3.7 Correlation Heatmap

The heatmap reveals which features correlate most strongly with attrition. Job Level, Monthly Income and Total Working Years show strong negative correlation — meaning higher values reduce attrition risk.

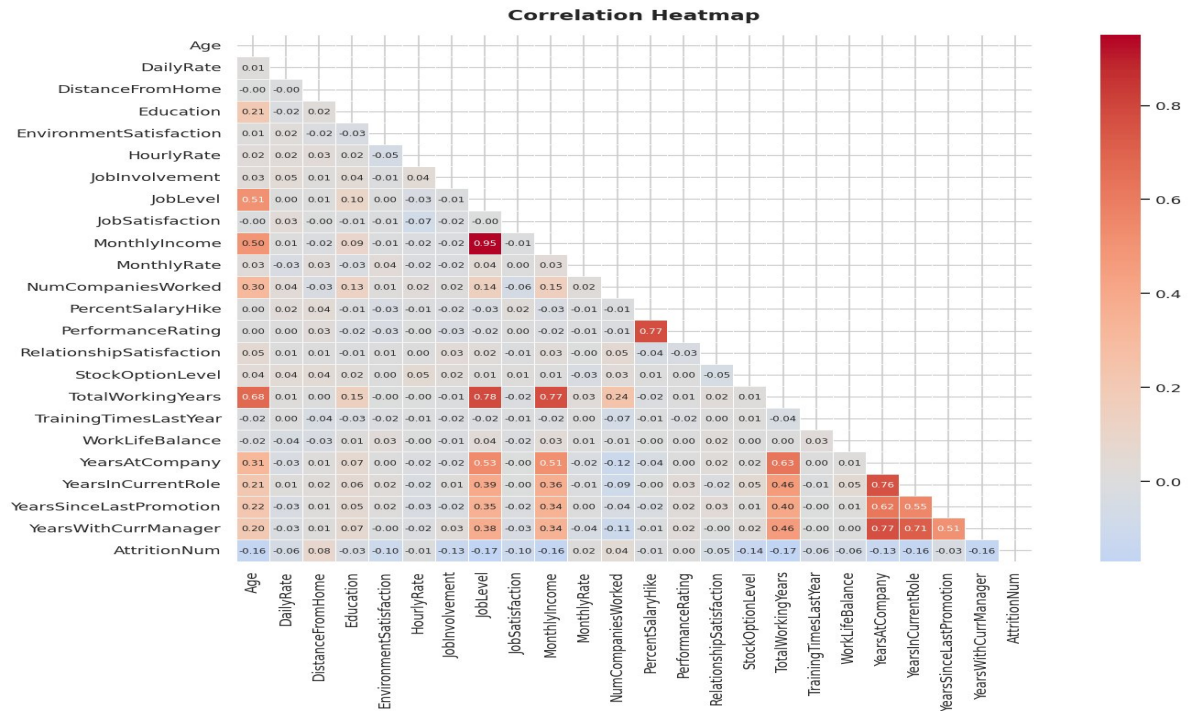


Figure 7: Correlation Heatmap of All Features

3.8 Job Satisfaction vs Attrition

Low job satisfaction (Level 1) leads to 22.8% attrition rate, while Very High satisfaction (Level 4) reduces it to 11.3%.

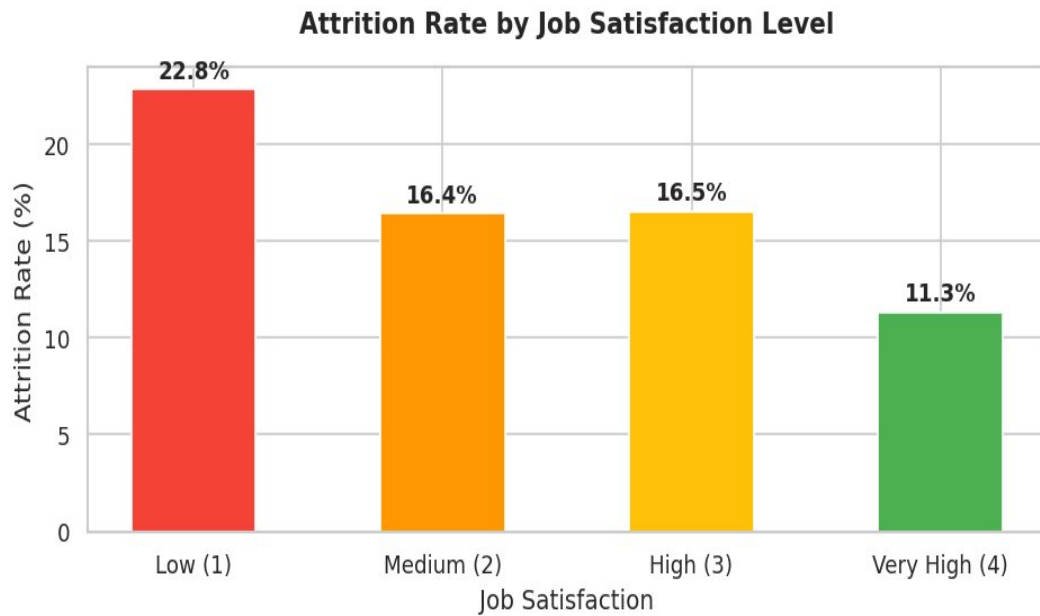


Figure 8: Attrition Rate by Job Satisfaction Level

4. Machine Learning Model Results

Three models were trained and evaluated on 80/20 train-test split with stratified sampling to handle class imbalance.

4.1 Model Performance Comparison

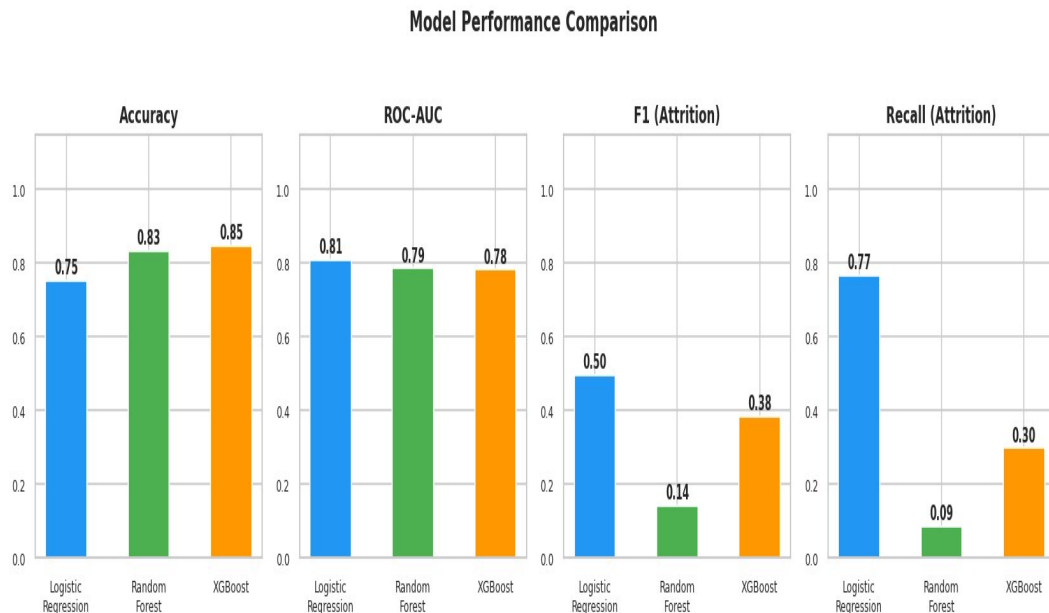


Figure 9: Model Performance Comparison — Accuracy, ROC-AUC, F1, Recall

4.2 ROC Curves

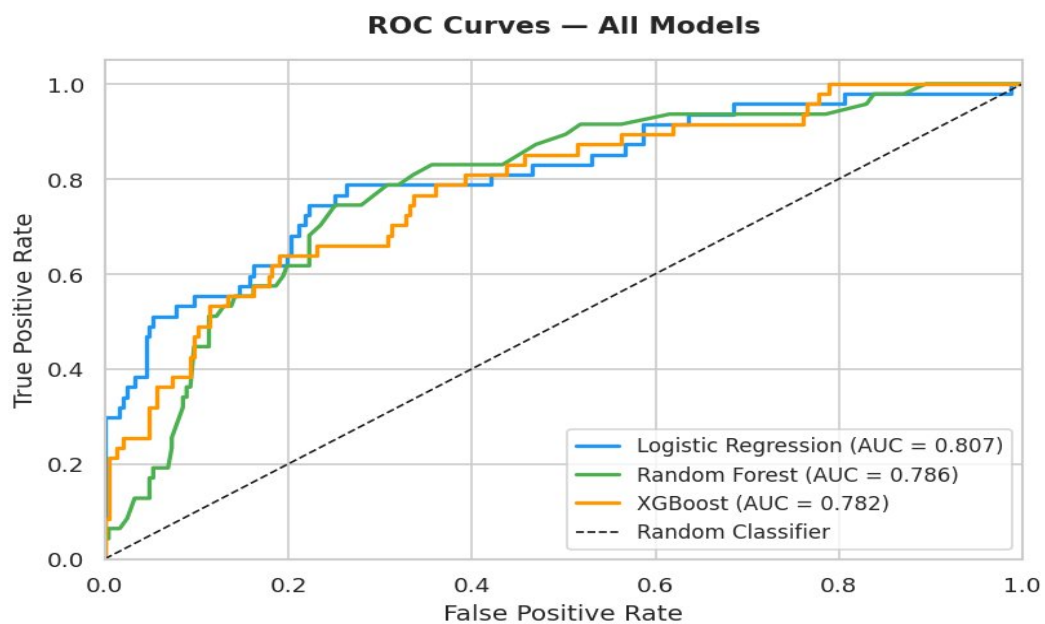


Figure 10: ROC Curves for All Three Models

4.3 Confusion Matrix — XGBoost

XGBoost correctly identified 235 employees who stayed and 14 who left. 33 at-risk employees were missed (false negatives).

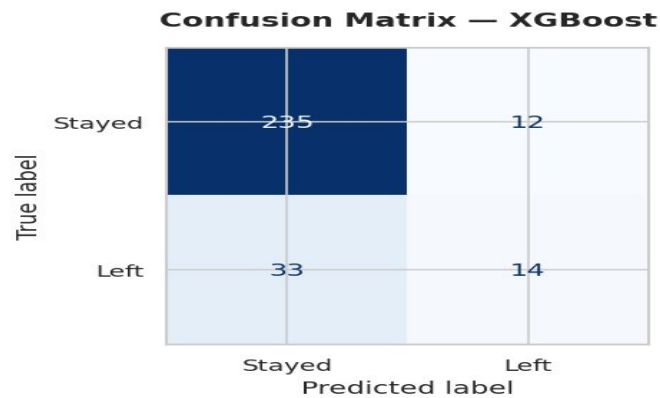


Figure 11: Confusion Matrix — XGBoost

Model	Accuracy	ROC-AUC	F1	Recall
Logistic Regression	75%	0.807	0.50	0.77
Random Forest	83%	0.786	0.14	0.09
XGBoost	85%	0.782	0.38	0.30

Logistic Regression achieved the best ROC-AUC (0.807) and highest recall (0.77) for identifying at-risk employees. XGBoost achieved the best overall accuracy (85%).

5. Feature Importance

Random Forest feature importance reveals Monthly Income as the strongest predictor of attrition, followed by Age, Daily Rate, and Total Working Years.

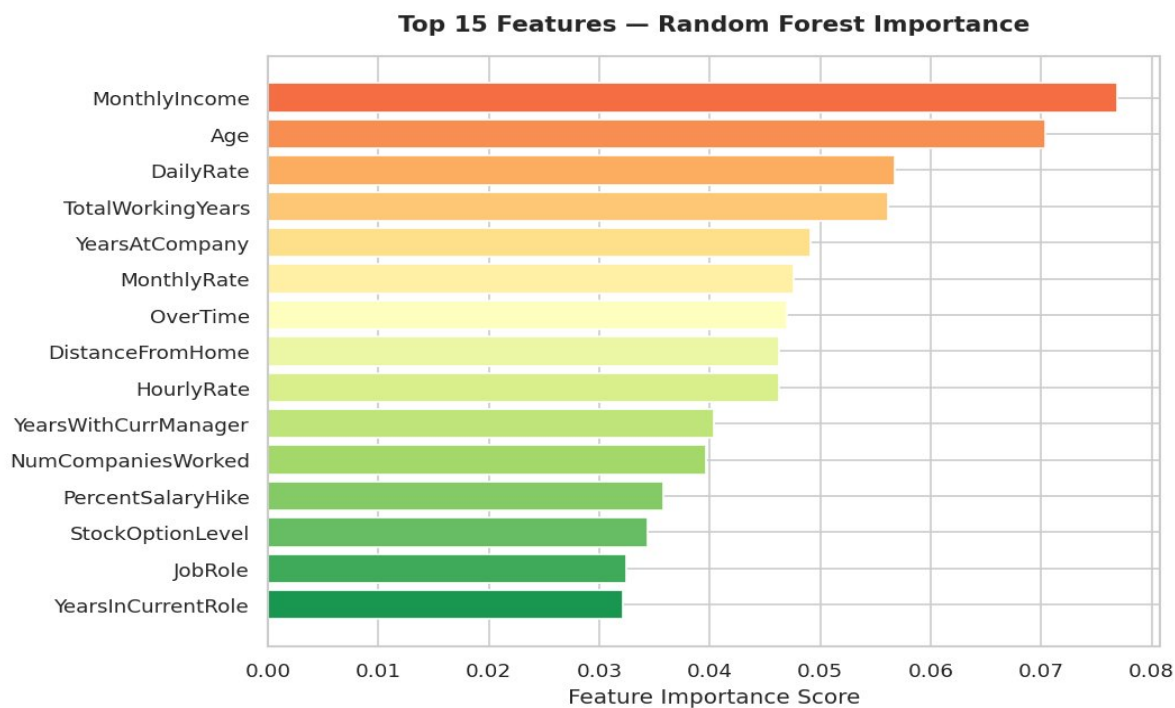


Figure 12: Top 15 Features by Random Forest Importance Score

6. Key Findings

#	Finding	Value
1	Overall attrition rate	16.1%
2	Highest risk age group	18-25 years (34.8%)
3	Highest risk department	Sales (20.6%)
4	Highest risk job role	Sales Representative (39.8%)
5	Overtime employees attrition	30.5% vs 10.4% (3x higher)
6	Low satisfaction attrition	22.8% vs 11.3% (Very High)
7	Top attrition predictor	Monthly Income
8	Best model recall	Logistic Regression (77%)

7. Business Recommendations

#	Recommendation	Target
1	Revise salary bands for low-income roles	Sales, Entry-level
2	Limit compulsory overtime policies	All departments
3	Launch retention programs for Sales department	Sales team
4	Build career growth pathways for young employees	Age 18-25
5	Deploy Logistic Regression model for early warning system	HR team
6	Conduct regular job satisfaction surveys	All employees

8. Conclusion

This project successfully demonstrated an end-to-end data analysis and machine learning pipeline on a real-world HR dataset. Key findings reveal that overtime, low monthly income, young age group, and poor job satisfaction are the primary drivers of employee attrition. Logistic Regression achieved the best recall (0.77) for identifying at-risk employees with ROC-AUC of 0.807, while XGBoost achieved the best overall accuracy of 85%.

The project covers the complete data science workflow — from raw data exploration to business recommendations — demonstrating practical skills in Python, statistical analysis, machine learning, and data visualisation.

Project Repository: github.com/jhanavirr-126/employee-attrition-analysis